

BORTNAN, Sweden

WMO: 023190

Lat: 62.7548N

Lon: 13.8427E

Elev: 467

StdP: 95.84

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.4	(c) -26.3	(d) -33.1	(e) 0.2	(f) -28.5	(g) -29.6	(h) 0.3	(i) -26.1	(j) 6.9	(k) 0.5	(l) 6.2	(m) -0.2	(n) 0.7	(o) 290	(p) 0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	24.3	15.0	22.5	14.8	20.4	13.8	16.7	21.6	15.7	20.1	14.6	18.5	2.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	11.2	18.0	13.8	10.4	17.0	12.7	9.7	16.2	48.6	21.6	45.4	20.0	42.2	18.5	20.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.2	(b) 5.3	(c) 4.7	(e) -33.1	(f) 26.6	(g) 2.6	(h) 2.0	(i) -34.9	(j) 28.0	(k) -36.4	(l) 29.2	(m) -37.8	(n) 30.3	(o) -39.7	(p) 31.8		
(a) 6.2	(b) 5.3	(c) 4.7	(e) -33.2	(f) 17.8	(g) 2.5	(h) 1.2	(i) -35.0	(j) 18.7	(k) -36.5	(l) 19.4	(m) -37.9	(n) 20.1	(o) -39.7	(p) 21.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-9.5	-8.3	-4.6	0.9	5.9	10.1	12.9	11.1	7.0	1.4	-4.1	-8.1
	DBStd	9.21	7.45	6.80	5.63	3.63	3.72	3.22	3.02	2.88	3.15	3.96	5.25	7.48
	HDD10.0	3394	604	512	452	272	137	39	8	22	97	268	423	562
	HDD18.3	6226	862	745	710	522	384	248	171	226	339	525	673	820
	CDD10.0	211	0	0	0	0	10	41	97	54	8	1	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	94	0	0	0	0	5	23	57	9	0	0	0	0
	CDH26.7	6	0	0	0	0	0	1	5	0	0	0	0	0
Wind	WSAvg	1.8	1.6	1.7	2.1	2.0	2.2	2.2	1.9	1.7	1.8	1.8	1.5	1.5
Precipitation	PrecAvg	638	38	29	30	34	43	78	100	87	61	54	45	38
	PrecMax	872	70	49	53	61	93	140	208	139	136	144	112	84
	PrecMin	503	4	3	6	8	14	18	23	46	14	9	14	11
	PrecStd	99	18	12	12	17	20	33	52	29	33	34	20	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	5.9	8.9	16.1	24.0	25.6	27.2	24.8	19.6	14.1	9.3	6.2
		MCWB	2.9	3.0	4.1	8.0	13.1	14.3	17.3	16.1	13.0	10.3	7.4	4.0
	2%	DB	3.8	4.5	6.7	13.2	20.9	23.7	24.9	22.4	17.3	11.5	6.2	4.6
		MCWB	2.0	2.0	3.5	6.8	12.1	13.5	16.5	15.3	12.3	8.5	5.1	2.8
	5%	DB	2.1	3.2	5.0	10.6	17.8	20.9	22.9	20.2	15.1	9.7	4.3	3.4
		MCWB	0.5	1.2	2.1	5.5	10.5	13.0	15.6	14.4	11.5	7.2	3.1	1.9
	10%	DB	0.4	1.3	3.5	8.3	14.6	18.2	20.7	18.1	13.4	8.1	2.8	1.5
		MCWB	-0.8	-0.4	1.0	4.1	8.8	11.7	14.5	13.6	10.3	6.4	1.7	0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.4	4.9	9.3	13.7	16.4	18.5	17.6	14.5	11.2	7.6	4.4
		MCDB	5.0	5.4	7.6	15.1	20.8	21.8	23.8	21.4	17.6	13.3	9.2	5.8
	2%	WB	2.0	2.3	3.7	7.1	12.2	14.7	17.2	16.3	13.0	8.9	5.1	3.0
		MCDB	3.7	4.2	6.9	12.3	17.8	20.3	22.8	20.2	16.1	10.7	6.1	4.4
	5%	WB	0.7	1.2	2.4	5.8	10.8	13.6	16.1	15.3	11.8	7.7	3.2	1.8
		MCDB	1.9	3.2	4.9	9.9	15.7	19.5	21.2	19.0	14.7	9.3	4.1	3.2
	10%	WB	-0.8	-0.2	1.3	4.5	9.1	12.5	15.0	14.2	10.7	6.4	1.8	0.3
		MCDB	0.3	1.4	3.4	8.0	13.3	17.7	19.1	17.6	13.0	8.1	2.6	1.3
Mean Daily Temperature Range	5% DB	MDBR	9.3	10.5	11.3	11.1	12.1	12.0	11.9	11.8	10.4	8.8	7.7	8.8
		MCDBR	8.6	8.6	10.8	16.0	18.9	18.1	16.7	16.3	14.3	11.9	8.3	8.9
		MCWBR	7.5	6.9	7.8	9.9	10.2	9.3	8.9	9.8	9.7	8.9	7.2	7.9
	5% WB	MCDBR	8.6	8.3	10.0	14.1	15.5	15.1	14.7	13.7	12.5	10.8	8.0	9.1
		MCWBR	7.6	6.8	7.4	9.0	9.3	8.3	8.4	8.8	8.9	8.5	7.1	8.2
Clear-Sky Solar Irradiance	taub	0.200	0.258	0.287	0.320	0.329	0.325	0.340	0.329	0.299	0.284	0.213	N/A	
	taud	2.077	2.271	2.318	2.394	2.477	2.518	2.489	2.507	2.574	2.489	2.151	N/A	
	Ebn at Noon	484	712	826	869	890	900	874	854	821	688	458	N/A	
	Edh at Noon	30	57	81	94	94	92	93	84	64	47	27	N/A	
All-Sky Solar Radiation	RadAvg	0.21	0.80	2.24	4.01	4.69	5.12	4.73	3.62	2.25	0.98	0.29	0.10	
	RadStd	0.02	0.12	0.21	0.33	0.39	0.43	0.56	0.33	0.24	0.14	0.05	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page